Type 1: Research Project

# Porygon: Learning Normal Behaviour Inside Running Containers

*Does accounting for how a container is deployed yield a more reliable notion of normal behaviour?*

**23CSE498 – Project Phase 2 (Panel Review 1)** | **Team Number: [XX]**


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Guide: **Guide Name**, Designation, Department of CSE


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# The Problem, in Plain Words

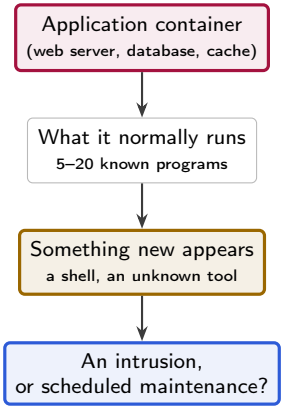
## Why watching a container is harder than it sounds

A **container** is a packaged application that runs isolated on a server. Companies run thousands of them.

If an attacker gets inside one, they run **programs** that the application would never normally run — a command shell, a download tool, a scanner.

So the natural idea is: *learn what programs a container normally runs, then flag the unusual ones.*

**The difficulty.** Before comparing against "normal", one must decide *which* normal. Grouping unlike containers together makes the comparison meaningless.




Frequently the correct answer is that the evidence is insufficient. The design reports that state explicitly rather than concealing it.

# What We Are Actually Asking

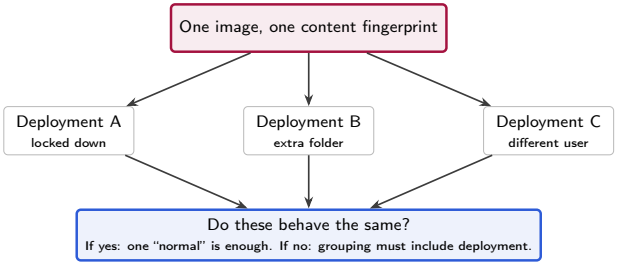
## The research question, and why it is not obvious

Every container image has a **content fingerprint** — a code computed from the exact bytes of the image. Same code, same program; it cannot be faked or quietly changed.

One could group containers by fingerprint and learn a single normal for each. Is that sufficient?

**Not entirely.** The same image can be *deployed* differently: another start-up command, user account, permission set or attached folder. Those differences may change behaviour.

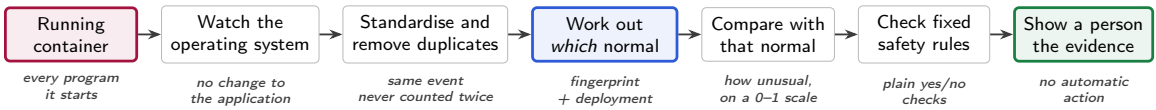
**Our question:** does adding *how it was deployed* produce a cleaner normal, or merely divide the data into groups too small to learn from?




Four grouping strategies are compared and the measurements decide the outcome. A result of "no difference" is a valid finding, not a failure.

# 1. Methodology — How the System Works, End to End

Rubric: Workflow & Algorithmic Pipeline (5 Marks)




**Two kinds of evidence, deliberately kept separate**

- **Statistical:** "this is rare compared with known-normal runs"
- **Rule-based:** "a command shell started in a web server"

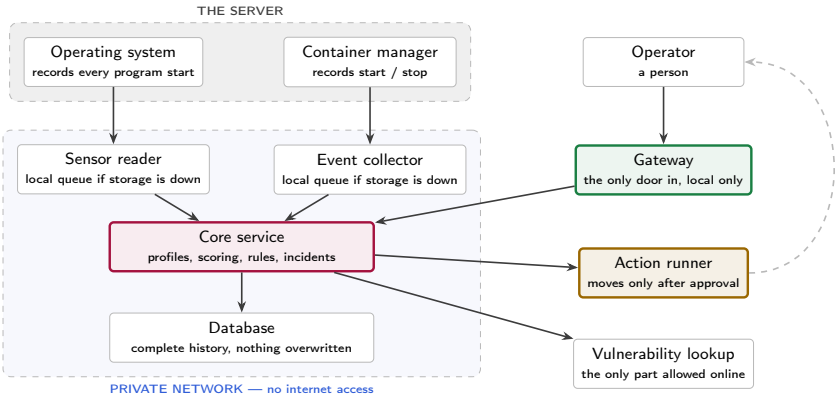
Combining them into a single figure would conceal which of the two applied.

**Deliberately excluded from the design**

- No automatic shutdown of containers
- No opaque machine-learning model
- No claim that unusual behaviour indicates an attack

# 1. Methodology — System Architecture

Rubric: Architecture & Trust Boundaries (5 Marks)




Each part gets the **least** access it needs. The core service never reaches the internet; the vulnerability lookup never touches the database; the action runner only moves when a person has approved.

# 1. Methodology — The Mathematics

Rubric: Mathematical Model / Formulation (5 Marks)

**Step 1 — How far has the mix of programs shifted?**
Compare the proportions seen now ($q$) with normal ($p$):

$$H(p,q) = \sqrt{\tfrac{1}{2}\sum_i \left(\sqrt{p_i} - \sqrt{q_i}\right)^2}$$

Always between 0 and 1. Zero means identical.

**Step 2 — How surprising is the order?**
Programs start in habitual sequences. Score each step by how unexpected it is:

$$S = -\tfrac{1}{n}\sum \log \hat{P}(\text{next} \mid \text{current})$$

Unobserved pairs retain a small non-zero probability, so no term divides by zero.

**Step 3 — How much is new?**
The proportion of activity whose program never appeared during learning.

**Step 4 — Combine without arbitrary weights**
Each measure is ranked against known-normal runs and the ranks are **averaged equally**. A measure that cannot be computed is recorded as unavailable, never estimated.

**Step 5 — Express the result on an interpretable scale**
Using $n$ complete known-normal runs withheld from learning:

$$p = \frac{1 + \#\{\text{normal runs at least this odd}\}}{n+1}$$

**Interpretation:** approximately one in twenty normal runs appears this unusual. This is *not* a probability of attack.

**Why no tuned weights?** Earlier fixed constants (0.50/0.30/0.20) had no justification, so the implementation now accepts none.

# 1. Methodology — Data, Splits and Evaluation Metrics

Rubric: Datasets, Preprocessing & Evaluation Metrics (5 Marks)

**The data is generated, not downloaded.**
No public dataset matches this question, so we produce our own under controlled conditions: three real, widely used server programs, each locked to an exact image fingerprint.

| Program | Locked version |
|---|---|
| Web server | nginx 1.26.3 |
| In-memory store | Redis 7.2.7 |
| Database | PostgreSQL 16.6 |

Each is run idle, under steady load, under bursts, and with deployment changes. Harmless but unusual activity (backups, restarts, maintenance) is recorded too: these are the cases that generate false alarms.

**Splitting rule:** a *whole run* goes to learning, calibration or testing, never divided. Observations seconds apart are near-duplicates; treating them as independent would overstate confidence.

**What we will measure**

| Group | Measures |
|---|---|
| Getting it right | false-alarm rate on harmless runs; how many planted events are caught |
| Trustworthy data | events created vs. events stored, at every hand-off point |
| Speed | delay from event to stored record (typical, slow, worst) |
| Cost | processor time, memory, disk growth, slowdown of the watched app |
| Honesty of scale | do 95% of normal runs really score inside the 95% band? |

**Reporting standard:** every percentage is published with the counts that produced it. No figure appears without the runs behind it.

# 2. Implementation Progress — Module Status

Rubric: Core Modules Implemented, Functional & Integrated (8 Marks)

| Module | What it does today | Status |
|---|---|---|
| System-level watcher | Every program start, in every container | **Working** |
| Container event collector | Start/stop events; resolves the exact image identity | **Working** |
| Safe delivery | Local queue; never skips an event it failed to store | **Working** |
| Storage & history | 9 schema upgrades; nothing overwritten | **Working** |
| Deployment fingerprint | Turns a deployment into one repeatable identity | **Working** |
| Behaviour profiles | Builds and versions "normal"; refuses weak ones | **Working** |
| Unusualness scoring | Explainable score with its reasons attached | **Working** |
| Rule-based evidence | Fixed checks, incident timeline, human sign-off | **Working** |
| Approved action runner | Pause or stop, only after a person approves | **Working** |
| Vulnerability lookup | Software inventory per image; known-exploited flags | **Working** |
| Experiment harness | Runs real containers; tamper-evident results | **Working** |
| Four-way comparison | The core research comparison | **Not started** |
| Result tables, figures | Cannot exist before the study runs | **Not started** |

**Overall ≈65%** — platform ≈90% complete, evaluation ≈15%. The split is reported rather than a single figure.

# 2. Implementation Progress — Evidence of Working Software

Rubric: Functional & Integrated (8 Marks)

**Size and coverage**

| | |
|---|---|
| Lines of program code | 17,802 |
| Automated tests passing | **156** |
| Service interfaces | 64 |
| Database upgrade steps | 9 |
| Services running together | 8 |

**Automatic checks, all passing**

| | |
|---|---|
| Code & configuration | **pass** |
| All unit tests | **pass** |
| Live end-to-end run | **pass** |
| Real-container run | **pass** |

**Measured on real containers (16 runs)**

| | |
|---|---|
| Marked test events created | 96 |
| Seen by the system watcher | **96** |
| Stored in the database | **96** |
| Lost | **0** |
| Counted twice | **0** |
| Application requests served | 360/360 |
| Leftover containers after cleanup | **0** |

**Complete chain proven on real data**


From a real database container: 479 recorded program starts became a profile, a later window scored *normal-looking*, and 72 rule checks matched without raising a false incident.

A system of this kind is only trustworthy if it declines at the correct moments. Three such refusals can be demonstrated live:

**1. It refuses a weak profile.**
Asked to learn "normal" from too little data, it stops:

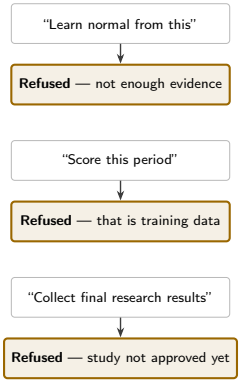
```
needed 3 time windows, found 2
```

**2. It refuses to evaluate itself on its own training data.**
Asked to score a period it already learned from:

```
scoring windows must not overlap the training interval
```

**3. It refuses an unstable reference.**
Asked to use a version label instead of an exact fingerprint:

```
refusing a mutable image reference
```




Each refusal is enforced in code and covered by an automated test, so it cannot be bypassed inadvertently.

# 3. Technical Knowledge — Key Decisions and Why

Rubric: Understanding of Methodology, Implementation & Design Choices (5 Marks)

| Decision | Why | What we rejected |
|---|---|---|
| Identify images by content fingerprint | A version label can be pointed at different software tomorrow; a fingerprint cannot | Grouping by container name or version label |
| Watch at the operating-system level | Sees every program start without changing the application | Modifying applications, or trusting logs the attacker can edit |
| Use a maintained sensor | Equivalent evidence at far lower risk than writing new system-level code | Building a custom system-level sensor: high effort, wide exposure |
| Combine measures by equal average | No number we cannot justify to an examiner | Weights tuned until the results look good |
| Judge against held-back normal runs | Turns a raw distance into "how often does normal look this odd?" | Declaring a threshold and calling it accuracy |
| Split by whole runs | Snapshots seconds apart are near-copies; reusing them fakes confidence | Random splitting of time windows |
| A person approves every action | A false alarm must never take down a healthy service | Automatic shutdown on a score |

# 3. Technical Knowledge — Trade-offs We Accepted

Rubric: Design Choices & Trade-offs (5 Marks)

| Benefit | Cost |
|---|---|
| Strong, cheap signal from program starts | Blind to file access and network destinations |
| Cleaner "normal" per deployment | Fewer runs per group; some groups too small to judge |
| No unjustifiable tuned numbers | Almost certainly not the most accurate possible |
| Honest confidence intervals | Need far more runs for the same certainty |
| A fair, interpretable scale | Only valid while the workload does not drift |
| No self-inflicted outages | Slower response to a real attack |
| One language across all services | A known write-speed ceiling, not yet a bottleneck |

**Edge cases already handled**

- Storage full → queue locally, never skip an event
- Sensor restarts → counters do not go backwards
- Same event twice → stored once
- Malformed input → kept as a short, scrubbed sample
- Too little data → answer "insufficient", never guess
- Secret on a command line → replaced before storage

# 3. Technical Knowledge — Early Observations

Rubric: Preliminary Observations (5 Marks) — measured, not expected

**1. No measured data loss to date.**
96 marked events created, 96 observed by the system watcher, 96 stored. Points that cannot yet be measured are labelled as such, never assumed to be zero.

**2. A defect that only repeated runs could reveal.**
The deployment identity accidentally included a name that changes every run, so the same deployment looked different each time. Left unfixed, every group would have stayed permanently "too small to judge" — and it would have looked like a finding rather than a defect.

**3. Serving traffic starts no new programs.**
40 web requests and 40 database-cache operations produced *zero* recorded program starts. The profile therefore describes start-up and maintenance activity, a limitation stated here rather than left implicit.

**4. The deployment changes we tested made no behavioural difference.**

| Program | Variants | Events each | Same? |
|---|---|---|---|
| Web server | 3 | 46 / 46 / 46 | yes |
| Database | 3 | 166 / 166 / 166 | yes |
| Cache | 3 | 33 / 33 / 33 | yes |

Restricting permissions and adding a scratch folder changed the *identity* but not one program that ran.

**Why this matters now:** running the full study with these settings would have produced a "no benefit" answer caused by our own choice of test conditions — not by the idea being wrong. We will switch to deployment changes that actually alter which programs run, *before* the study is locked.

Identifying this before the study is locked is the purpose of a trial run.

# 3. Technical Knowledge — What We Do Not Claim

Knowing the limits is part of understanding the method

**Unusual is not the same as malicious.**

- Unusual but harmless: backups, debugging, restarts, a traffic spike
- Harmful but ordinary-looking: an attacker using tools already present

We keep the two ideas in separate fields and never merge them into one verdict.

**What the system can miss**

- Anything that is not a program start
- Slow activity spread thinly over time
- Behaviour that was already present while learning
- Any container whose group has too few examples

**Not yet claimed, and why**

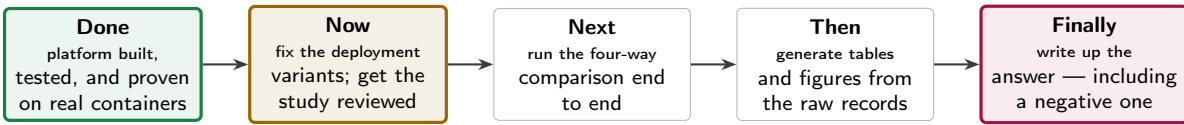
| Claim | Blocked by |
|---|---|
| Detection accuracy | final study not run |
| False-alarm rate | final study not run |
| Which grouping wins | the comparison itself |
| Suitable for production | single machine, research scope |

**Current blockers**

1. Study design awaiting two independent human reviews — the code refuses to collect final results until then
2. 65 GB free disk against 150 GB needed for the full study
3. Deployment variants need revising (previous slide)

# Where This Goes Next

From a working platform to an answered question




**The intended contribution** is not container monitoring itself, which is well-established work. It is a controlled, repeatable comparison of *four different definitions of what counts as the same thing*, reported faithfully whichever way the result falls.

Every figure will be traceable to the run that produced it, and every run is retained so that a reviewer can recompute the tables independently.

- **Individual MOOC / Online Certification Selection:**

| S.No | Student Name | Chosen MOOC / Course Title | Platform | Hours |
|---|---|---|---|---|
| 1 | Anurup R Krishnan | [e.g. Container Security / Linux Observability] | Coursera | 15 Hrs |
| 2 | Student Name 2 | [e.g. Applied Statistics for Experiments] | NPTEL | 20 Hrs |
| 3 | Student Name 3 | [e.g. Docker & Cloud-Native Fundamentals] | edX / Coursera | 12 Hrs |
| 4 | Student Name 4 | [e.g. Cloud Security Foundations] | AWS / Google | 16 Hrs |

- **Requirement:** Each student must complete one approved 10–20 hour online certification from a recognized platform (Coursera, NPTEL, edX, etc.) relevant to the project.
- **Note:** course titles above are suggestions aligned to the project's four skill areas — container internals, operating-system observation, experiment design, and security practice. Replace with the actual enrolled courses and attach completion certificates.

# References I


S. Forrest, S. A. Hofmeyr, A. Somayaji, and T. A. Longstaff, "A sense of self for Unix processes," in *Proc. IEEE Symp. Security and Privacy*, 1996, pp. 120–128.

S. A. Hofmeyr, S. Forrest, and A. Somayaji, "Intrusion detection using sequences of system calls," *Journal of Computer Security*, vol. 6, no. 3, pp. 151–180, 1998.

A. S. Abed, T. C. Clancy, and D. S. Levy, "Applying bag of system calls for anomalous behavior detection of applications in Linux containers," in *Proc. IEEE Globecom Workshops*, 2015, pp. 1–5.

J. Lin, "Divergence measures based on the Shannon entropy," *IEEE Trans. Information Theory*, vol. 37, no. 1, pp. 145–151, 1991.

V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York, NY, USA: Springer, 2005.

G. Shafer and V. Vovk, "A tutorial on conformal prediction," *Journal of Machine Learning Research*, vol. 9, pp. 371–421, 2008.

B. Gregg, *BPF Performance Tools: Linux System and Application Observability*. Boston, MA, USA: Addison-Wesley, 2019.

The Falco Authors, "Falco: cloud-native runtime security," Cloud Native Computing Foundation. [Online]. Available: https://falco.org

# References II


Open Container Initiative, “OCI image format specification.” [Online]. Available: https://github.com/opencontainers/image-spec

# Thank You

Questions welcome, including those on the limits of what this system can currently demonstrate.